

Pratheek Shetty

Technical Lead — Distributed Systems | Backend Platforms | Real-Time Data Infrastructure

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PROFESSIONAL SUMMARY

Technical Lead with 6+ years of experience in System Design, engineering large-scale distributed systems, real-time data platforms, and cloud-native infrastructure. Expert in microservices architecture, event-driven stream processing, IoT communication systems, and high-availability backend platforms leveraging Kafka, Apache Flink, MQTT, WebSockets, Apache Pinot, PostgreSQL, and AWS/GCP ecosystems. Currently leading cross-functional engineering teams at Simple Energy, architecting data pipelines that process billions of telemetry events while strategically optimizing infrastructure to reduce cloud expenditure by 44%. Proven track record establishing core software patterns, driving strict technical execution, and taking platform products seamlessly from initial design to planet-scale production.

CORE EXPERTISE

System Design & Architecture

Distributed Systems Infrastructure

Real-Time Ingestion & WebSockets

Relational Databases & PostgreSQL

Event-Driven Stream Processing

Infrastructure Cost Optimization

Cross-Functional Engineering Leadership

Connected Vehicle & IoT Platforms

Generative AI & Data Lakes

CRM Integration & CDC Pipelines

TECHNICAL SKILLS

Data & Streaming	Kafka, Apache Flink, Apache Spark, EMQX MQTT Broker, MQTT Protocol, WebSockets, Apache Pinot, ClickHouse, Redis, PostgreSQL, SQL, Delta Lake, Amazon Redshift, AWS DMS, Storage Tiering
Backend & Languages	Node.js, TypeScript, JavaScript, Express.js, GraphQL, REST APIs, Electron.js, LangGraph, LLM Orchestration, AI Support Automation, System Design, SOLID Principles
Frontend Platform	React.js, Next.js, Redux, D3.js, Micro-Frontends, Responsive UI Architecture, Material UI
Cloud & Infrastructure	AWS (EC2, ECS, Lambda, Cognito, IoT Core, S3, SNS, SQS, RDS, AppSync), GCP (Kubernetes/ GKE, Compute Engine, GCR, GCS), CI/CD, Git, Infrastructure as Code (IaC)

KEY INFRASTRUCTURE ACHIEVEMENTS

- Cost Optimization:** Reduced cloud infrastructure expenditure by 44% through data telemetry optimization, storage tiering, resource consolidation, and architectural improvements.
- Scale Engineering:** Architected connected vehicle telemetry pipelines processing billions of telemetry data points, maintaining 500B+ analytical data records in Apache Pinot for sub-second fleet analytics.
- Notification Throughput:** Built event-driven notification infrastructure utilizing WebSockets and Pub/Sub systems capable of handling millions of critical alert notifications per day at sub-second latencies.
- Team Leadership:** Led and coordinated a 23-member cross-functional engineering team spanning backend, frontend, mobile apps, DevOps, data engineering, and data science functions.

PROFESSIONAL EXPERIENCE

Technical Lead — Simple Energy

Bengaluru, India | Mar 2022 – Present

Connected vehicle platform powering real-time telemetry, OTA updates, and analytics infrastructure for thousands of production electric vehicles.

- Platform Strategy & System Design:** Owned the complete architectural lifecycle and high-level system design of the connected vehicle platform, supporting large-scale telemetry ingestion and real-time analytics pipelines.
- Stream Processing & Ingestion:** Designed a dual-pipeline ingestion model utilizing Kafka, EMQX MQTT Broker, and WebSockets to execute sub-second telemetry analytics across streaming vehicle datasets.
- Database Management:** Engineered transactional data stores and relational schemas using PostgreSQL, ensuring optimized query performance, database normalization, and fault-tolerant storage tiering for high-volume writes.

- **OTA Platforms:** Engineered secure Over-the-Air (OTA) firmware delivery backend infrastructures enabling cryptographic validation and automated remote fleet updates.
- **AI Platforms & Systems:** Architected a production enterprise generative AI system utilizing Amazon Redshift, AWS DMS, and S3 to orchestrate organizational knowledge indexing, delivering context-aware natural language insights across corporate data lakes.
- **AI Support Orchestration:** Built custom telemetry intelligence and support automation agents using LangGraph and LLM orchestration frameworks for automated vehicle troubleshooting and analytics workflows.
- **Operational Tooling:** Created low-latency diagnostic tools with Electron.js, optimizing factory-floor automated testing, provisioning validation, and hardware configuration routines.
- **Integration Architecture:** Designed and implemented a low-latency business integration connector layer facilitating high-fidelity LMS and Salesforce integrations; orchestrated real-time CDC (Change Data Capture) pipelines to stream updates directly from transactional RDS layers to external CRM endpoints.
- **Engineering Standards:** Established design patterns across Node.js microservices, conducted regular architecture reviews, and instituted automated CI/CD safeguards to maximize service reliability.
- **Frontend Platform Engineering:** Led the frontend architecture strategy for mission-critical internal analytics, vehicle diagnostics dashboards, and management portals using React.js, Next.js, and Redux.

Software Engineer — Tech Active

Bengaluru, India | Oct 2021 – Feb 2022

- **Product Architecture:** Directed system design and implementation of backend APIs for an audio licensing and contract collaboration ecosystem leveraging Node.js and scalable database schemas.
- **Dispatch Integration:** Designed internal service management and real-time logistics tracking engines using Next.js and WebSockets, linking live driver status assignments with automated billing workflows.
- **Performance Tuning:** Optimized database indexing in PostgreSQL and server-side caching mechanics on multi-tenant REST APIs, stabilizing backend response times under heavy concurrent traffic loads.

Frontend Developer — Filmyscoop

Bengaluru, India | Apr 2020 – Sep 2021

- **IoT Integration:** Engineered an IoT-enabled smart lighting controller platform backend interfaces and web control applications from concept to full production deployment within a 2-month window.
- **UI Standardization:** Constructed modular, reusable interface components in React.js, focusing heavily on client-side state optimization and asset performance.

TECHNICAL WRITING & THOUGHT LEADERSHIP

Published technical deep-dives on distributed architectures, real-time analytics pipelines, and engine mechanics on Medium:

- **Building a Scalable Connected Vehicle Platform** with HMI, EMQX, Kafka, Flink, and Pinot.
- **Evolving Our Connected Vehicle Data Platform:** Smart Dual-Pipeline Architecture.
- **Apache Pinot Optimization** for Real-Time IoT Telemetry Analytics.
- **Building a Production-Grade Support AI Agent** using modern LangGraph structures.
- **Handling Millions of Notifications at Scale** using event-driven architectures.
- **The Tale of Node.js:** Deconstructing Call Stack execution mechanics and the Event Loop.

EDUCATION

Mangalore Institute of Technology & Engineering (MITE)

2016 – 2020

Bachelor of Technology in Computer Science and Engineering